NATIONAL RESEARCH UNIVERSITY

Spatial aspects of unemployment in Russia: what is more important, sectoral proximity or geographical proximity?

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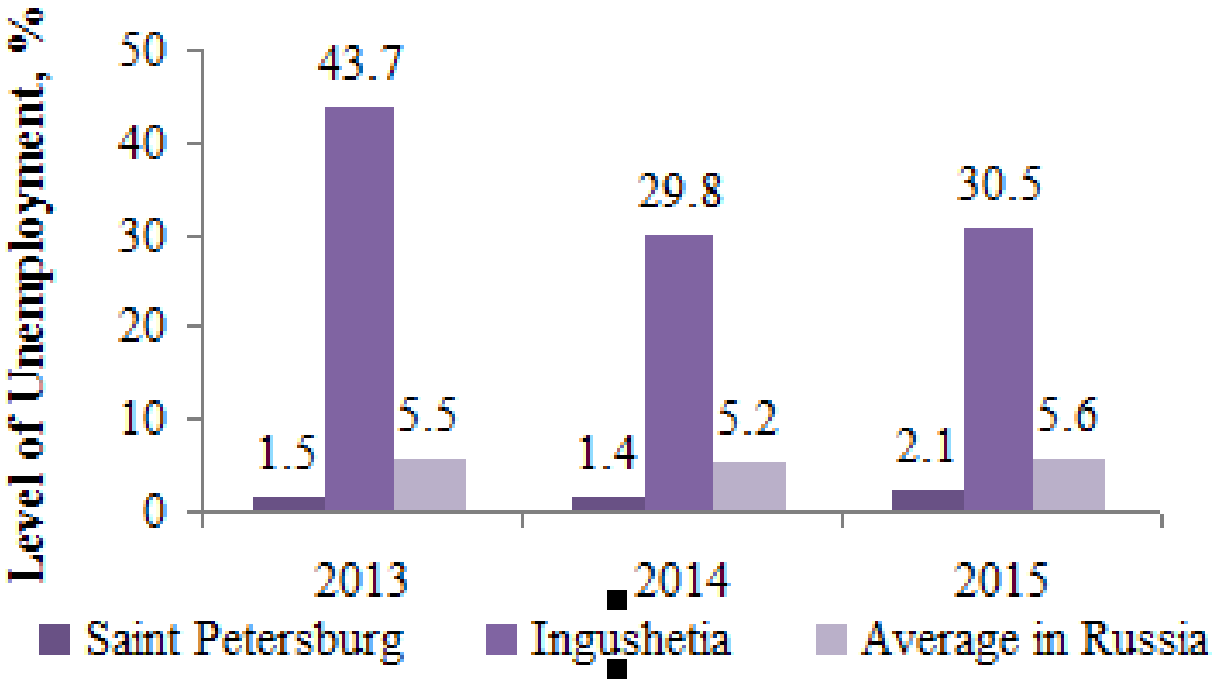
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Special features of Russian regional labour markets

- Huber (2007) and Bah and Brada (2014), A review of the papers devoted to the regional labour market in transition countries
- Pastore and Missuda (2015, introduction), “the Russian case seems to be specific and interesting not only among other transition countries but also in the European perspective”
- Kapelyushnikov et al. (2012): “the current model of labour relations in Russia is a combination of very formal rules embodied in the Labour Code and a great variety of informal arrangements that make it feasible to relax those rules
- Vakulenko and Gurvich (2016): “high wage flexibility is an important salient feature of the Russian labour market”

Special features of Russian regional labour markets

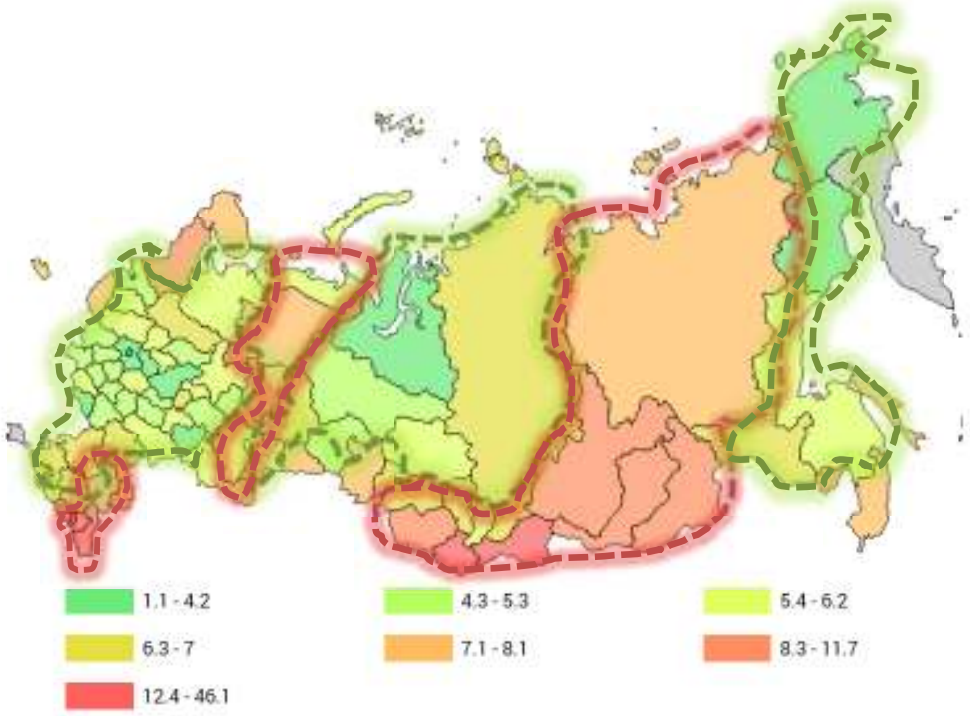
•Oschepkov and Kapelyushnikov , (2015) “the ratio between the minimum (1.4% in St. Petersburg) and the maximum (about 30% in Ingushetia) regional unemployment levels is over twenty” .



Special features of Russian regional labour markets

- Oschepkov and Kapelyushnikov (2015): “there is no single labour market in Russia, only a system of local labour markets”
- Russian regions are prone to clustering

Fig. Unemployment level in Russia in 2013



Motivation

Russia is rich in natural resources. However, these resources are located in a small number of regions.

•The aim of the research:

To determine whether other Russian regions receive spillovers from regions rich in natural resources

Spatial Models with Russian Regional Data

- **Economic Growth**

  Lugovoy et al. (2007), Solanko (2008), Ledyaeva et al. (2008), Kholodilin et al. (2012), Akhmedjonov et al. (2013), Lehmann and Silvagni (2013), Dolinskaya (2002), Kolomak (2011)

- **Migration**

  Sardadvar, Vakulenko (2016), Vakulenko (2015),

  Sardadvar, Vakulenko (2017), "A model of interregional migration under the presence of natural resources: theory and evidence from Russia"

- **Unemployment rate**

  Demidova and Signorelli (2012), Demidova et al. (2013), Demidova et al. (2015), Blinova et al. (2015), Blinova et al. (2016), Rusanovskiy and Markov (2016)

Spatial effects could not be neglected!

Weighting matrices

Conley, T. G., & Topa, G. (2002). Socio-economic distance and spatial patterns in unemployment

Two types of weighting matrices:

Based on the geographic proximity of regions

- Boundary
- Inverted distance

- Based on the proximity of the sectoral structures of gross value added (the Euclidean distance between 15-dimensional vectors reflecting the sectoral structure was used)

Weighting matrix based on the proximity of the gross value added by economic activity

$$\rho(\overrightarrow{x_i}, \overrightarrow{x_j}) = \|x_i - x_j\|_e = \sqrt{(x_{1i} - x_{1j})^2 + (x_{2i} - x_{2j})^2 + \cdots + (x_{15i} - x_{15j})^2}$$

	x_1	x_2	x_3	x_4	x_5	x_6			...						x_{15}
Белгородская область	100	11,6	0,0	21,8	23,1	3,3	6,6	13,6	0,6	6,5	0,0	3,9	2,9	2,7	2,5	0,9
Брянская область	100	14,3	0,0	0,1	21,7	4,8	3,8	15,7	1,4	15,3	0,0	6,4	5,9	4,5	5,1	1,0
Владимирская область	100	10,9	0,0	0,3	34,3	5,7	4,4	10,7	1,1	10,3	2,2	5,6	5,2	3,6	4,0	1,7

Gross value added by economic activity	
agriculture, forestry	information and communication
fishing	financial and insurance activities
mining and quarrying	real estate, rent and services activities
manufacturing	public administration and defense; compulsory social security
production and distribution of electricity, gas and water	education
construction	human health and social work activities
wholesale and retail trade; repair of motor vehicles and motorcycles	provision of other communal, social and personal services
accommodation and food service activities	

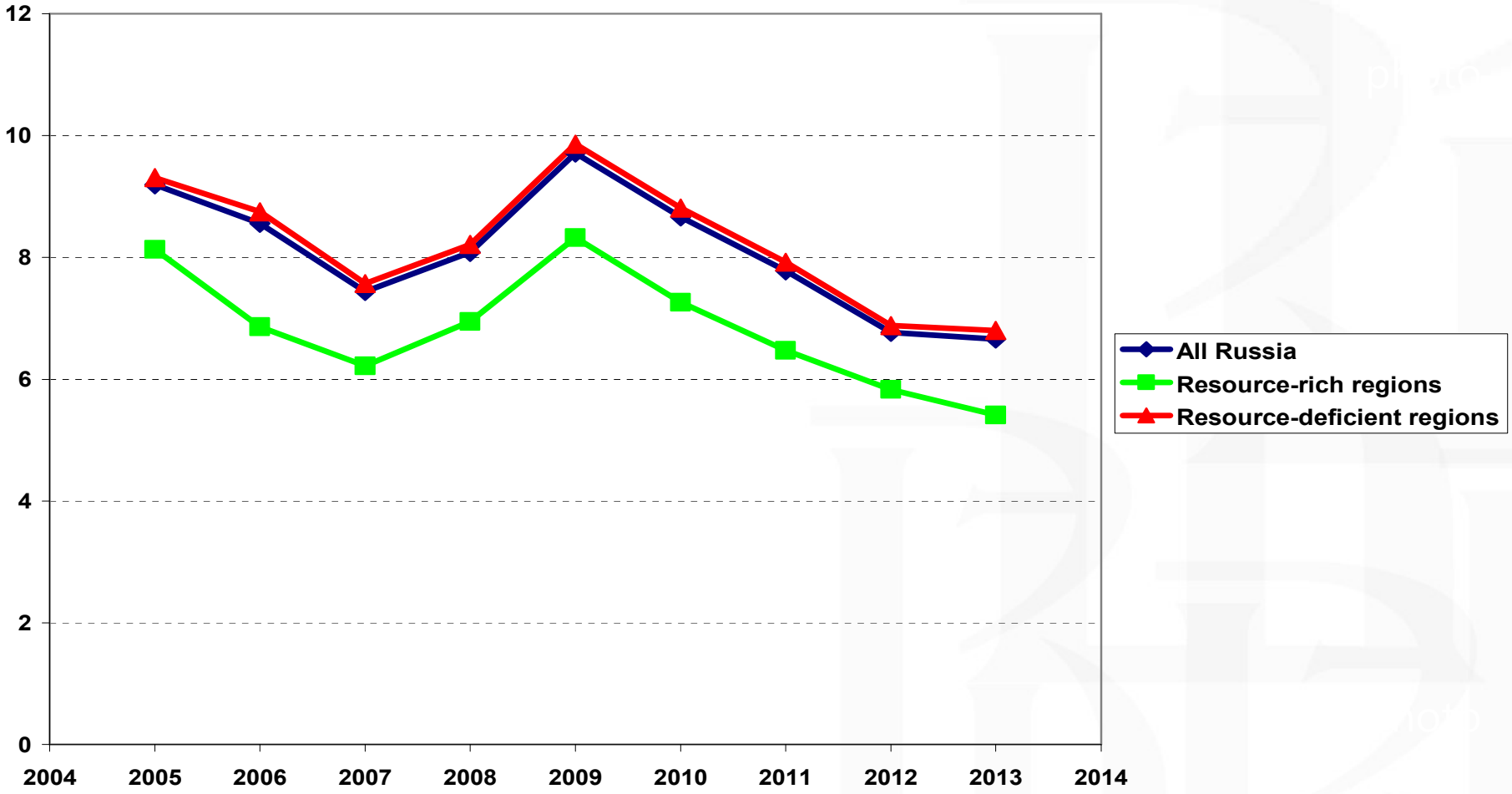
Resource-rich and resource-deficient regions

► The criterion for classifying the region as resource-rich was the share in the total volume of extraction of fuel and energy resources exceeding 3%.

► With this criterion, eight RF subjects were classified as resource-rich, including
the Republic of Tatarstan, the Orenburg Region, the Khanty-Mansi Autonomous Area-Yugra, the Yamal-Nenets Autonomous District, the Krasnoyarsk Territory, the Kemerovo Region, the Republic of Sakha (Yakutia) and the Sakhalin Region.

► The remaining 72 subjects were attributed to be resource-deficient (poor).

Dynamics of the average unemployment rate in Russia in 2005-2013



Descriptive statistics for unemployment

		Mean	Std.Dev.	Min	Max
All	overall	8.0989	5.923945	1.1	64.9
Russia	between		5.698865	2.222222	51.97778
	within		1.725519	-0.17888	21.02112
Resource	overall	6.831944	1.793794	2.7	10
rich regions	between		1.287622	4.433333	8.555555
	within		1.321568	4.020833	9.743056
Resource	overall	8.239673	6.200549	1.1	64.9
poor regions	between		5.980972	2.222222	51.97778
	within		1.765528	-0.0381	21.1619

Moran's Index

$$I(X) = \frac{N}{\sum_{i,j} w_{ij}} \frac{\sum_{i,j} w_{ij}(X_i - \bar{X})(X_j - \bar{X})}{\sum_{i} (X_i - \bar{X})^2}$$

Moran's Index

Year	Boundary Matrix			Inverted Distance Matrix		
	All Russia	Rich regions	Poor regions	All Russia	Rich regions	Poor regions
2005	0,078*	0,358***	0,153***	0,140***	0,398***	0,215***
2006	0,162***	0,321***	0,223***	0,223***	0,378***	0,285***
2007	0,162***	0,323***	0,232***	0,225***	0,377***	0,295***
2008	0,143***	0,339***	0,223***	0,207***	0,372***	0,285***
2009	0,071	0,376***	0,176***	0,098**	0,417***	0,206***
2010	0,061	0,337***	0,166***	0,079***	0,370***	0,193***
2011	0,053	0,315***	0,155***	0,066***	0,348***	0,175***
2012	0,03	0,320***	0,122**	0,057***	0,351***	0,153***
2013	0,056	0,335***	0,150***	0,093*	0,372***	0,189***

Main hypotheses

Hypothesis 1: There exist “geographical” spillovers in Russian labour market

Hypothesis 2: There exist structural-sectoral spillovers in Russian labour market

Data and Model

Data source: Federal State Statistics Service of the Russian Federation, www.gks.ru

80 Russian regions; period 2005 – 2013;
The dependent variable is regional unemployment rate

$$\begin{pmatrix} Y^r{}_{i_r t} \\ Y^p{}_{i_p t} \end{pmatrix} = \sigma \begin{pmatrix} Y^r{}_{i_r t-1} \\ Y^p{}_{i_p t-1} \end{pmatrix} + \begin{pmatrix} \rho_{rr} W_{rr} & \rho_{rp} W_{rp} \\ \rho_{pr} W_{pr} & \rho_{pp} W_{pp} \end{pmatrix} \begin{pmatrix} Y^r{}_{i_r t} \\ Y^p{}_{i_p t} \end{pmatrix} + \begin{pmatrix} X^r \beta_r \\ X^p \beta_p \end{pmatrix} +$$

$$+ \sum_{k=2007}^{2013} \gamma_k d_{200k} + \alpha_i + \varepsilon_{it}$$

Independent Variables

- GRP per capita
- population density
- share of urban population
- share of population under working age
- share of population over working age
- share of people with higher education in labour force
- migration growth of the population per 10,000 people

Methodology

- Weighting matries based on the proximity of the sectoral structure are endogenous

Kelejian, H. H., & Piras, G. (2014). Estimation of spatial models with endogenous weighting matrices, and an application to a demand model for cigarettes. Regional Science and Urban Economics, 46, 140-149.

Instruments for wij: distances between capitals of regions I and j, ratio of populations in regions I and j

- GMM

Kelejian & Piras approach

1) All nonzero elements of weighting matrices were instrumented

Instruments for wij: distances between capitals of regions i and j, ratio of populations in regions i and j

And their second and third powers.

2) Arellano-Bond approach, GMM

Results of estimation

Variables	Wb	Wid	Wend		Variables	Wb	Wid	Wend
Time lag	0.47***	0.575***	0.564***		youngr	-1.294	-3.865*	-2.617*
WYrr	0.411**	0.289***	-20.285*		youngp	-0.1	-0.233	0.126
WYpp	0.139*	0.037	-0.08		oldr	0.885	1.782	1.548*
WYrp	0.435***	0.323***	1.101		oldp	-0.047	-0.371	-0.465*
WYpr	0.691***	0.176	-0.405		higheducr	0.292	0.212*	-0.188
gdpr	-0.000	-0.000*	-0.00***		higheducp	0.01	-0.019	-0.015
gdpp	-0.000	-0.000	-0.000		migrat(-1)r	-0.002	-0.000	-0.000
densityr	-7.99	2.898	7.951		migrate(-1)p	-0.01***	-0.007***	0.001
densityp	0.004*	-0.001	-0.000		Time effects	Yes	Yes	Yes
urban sharer	-1.308	-2.89***	-1.997**		Sargan stat.	25.326	32.427	19.402
urban sharep	0.225**	0.148**	0.031		N	80	80	80

Conclusions

- All the hypotheses received empirical confirmation
- The existence of "geographic" spillovers was confirmed
- Positive "geographic" spillovers from resource-rich regions spread only to neighboring regions (i.e. if unemployment in resource-rich regions decreases, a similar change will occur in neighboring regions)
- “Structural-sectoral" spillovers were identified only for resource-rich regions and they were negative. This may indicate competition for labor resources between rich natural resources regions.

Thank you!

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